

# Online Book Store

## Requirement Analysis Assignment

### 1. Requirement Understanding

The Online Book Store is an e-commerce platform that allows customers to discover, evaluate, and purchase books online. The primary business goal is to provide users with a fast, secure, and intuitive shopping experience from browsing the catalog to completing a purchase while giving the business a reliable way to manage inventory, orders, and customer accounts.

The platform should support the following key user workflows:

- Users can **view books** organized by category, genre, or featured bestseller lists or by ratings.
- Users can **find specific books** using keywords such as title, author, and category.
- Users can **open a books detail page** to see its description, price, author, ratings reviews, and availability.
- Users can **add books** to a cart, update quantities, or remove items before checkout.
- Users can **check out**, enter shipping and payment details, and complete an order.
- Users can **register, log in, update** their profile, and view order history.

The system is expected to behave consistently across sessions (like cart contents persist for logged-in users), respond quickly to user actions, and protect sensitive data such as passwords and payment information.

### 2. Functional Requirements

Functional requirements define specific behaviors and features the system must provide.

| ID    | Requirement Description                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall allow a new user to register using name, email, phone number and password.                                       |
| FR-02 | The system shall allow a registered user to log in using valid credentials and log out of their session.                          |
| FR-03 | The system shall allow users to browse the book catalog by category, genre, or author.                                            |
| FR-04 | The system shall allow users to search for books by title, author, or category and display matching results.                      |
| FR-05 | The system shall display detailed information for a selected book, including title, author, price, description, and availability. |
| FR-06 | The system shall allow users to add one or more books to their shopping cart.                                                     |
| FR-07 | The system shall allow users to update quantities or remove items from the shopping cart.                                         |
| FR-08 | The system shall allow users to proceed to checkout, enter shipping details, and complete payment for items in the cart.          |
| FR-09 | The system should send an order confirmation on screen and through email once a purchase is successfully completed.               |
| FR-10 | The system should allow users to view their past order history from their account.                                                |

### 3. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints under which the system must operate.

| ID     | Requirement Description                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Book catalog and search result pages shall load within 3 seconds under normal network conditions.                                                         |
| NFR-02 | User passwords and payment details shall be encrypted in transit (HTTPS/TLS) and at rest.                                                                 |
| NFR-03 | The system shall maintain at least 99.5% uptime, excluding scheduled maintenance windows.                                                                 |
| NFR-04 | The interface shall be intuitive for first-time users and comply with basic web accessibility standards.                                                  |
| NFR-05 | The system shall support a growing catalog and concurrent user load (for example, 1,000+ users at same time) without significant performance degradation. |

### 4. User Stories

User stories describe system functionality from the perspective of different user roles.

- **As a** new visitor, **I want** to register for an account, **so that** I can save my details and track my orders.
- **As a** registered user, **I want** to log in securely, **so that** I can access my account and order history.
- **As a** shopper, **I want** to browse books by category, **so that** I can discover titles that interest me.
- **As a** shopper, **I want** to search for a book by title or author, **so that** I can quickly find a specific book I'm looking for.
- **As a** shopper, **I want** to view detailed information about a book, **so that** I can decide whether to purchase it.
- **As a** shopper, **I want** to add a book to my cart, **so that** I can purchase it along with other items later.
- **As a** shopper, **I want** to remove an item from my cart, **so that** I only pay for books I actually want.
- **As a** shopper, **I want** to check out and pay securely, **so that** I can complete my purchase with confidence.
- **As a** buyer, **I want** to receive an order confirmation, **so that** I know my purchase was successful.
- **As a** returning customer, **I want** to view my past orders, **so that** I can track deliveries or reorder books I liked.

### 5. Acceptance Criteria (Gherkin Syntax)

The following scenarios define testable acceptance criteria for key user stories, written using Given/When/Then syntax.

**Scenario: Successful user registration**

**Given** the user is on the registration page

**When** the user enters a valid name, unique email, and a valid password and submits the form

**Then** the system should create the account and redirect the user to the login page or dashboard

